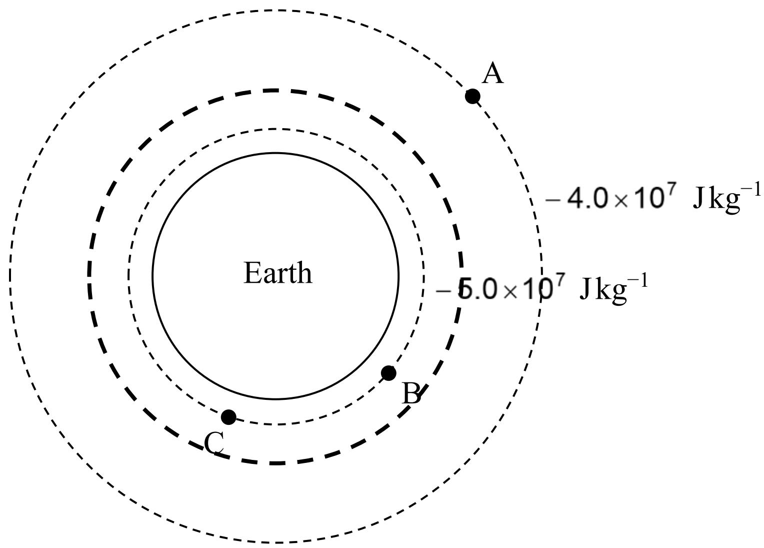
5(a)	Gravitational force is attractive in nature. Hence, the work done per unit mass, by an external force acting against the gravitational force is negative as the displacement and the external force are opposite in direction. Or Since gravitational potential at infinity is defined to be zero, and yet it increases with distance from the mass, its value has to be negative. <u>Comments:</u>	[A1] for any acceptable explanation

Work done by external force is poorly understood. Some students confuse unit mass with point mass.

5(b)(i)



[A1] for line drawn nearer to the -5.0×10^{7} J kg^{-1} equipotential line

Comments:
Large number of students did not ensure the line drawn is nearer to the -5.0 equipotential line.

5(b)(ii) Work done by gravitational field

$= m(\phi_A - \phi_B)$

$= 3000\times(-4.0+5.0)\times10^{7}$

$= 3.0\times10^{10}$ J

[M1] for correct substitution

[A1] for correct answer

Comments:
Many did not realise the question asked for work done by gravitational force, which is negative of the conventional GPE.

5(b)(iii) Since the force and displacement vectors are always perpendicular to each other, the work done is therefore zero.

[A1] for correct explanation [zero mark if mention B and C are at same potential]

Comments:
Expectedly, students tend to discuss equipotential values rather than use first principle like definition of work done.

5 (c) (i) The gravitational field strength g is not constant [B1]. The student's value would be greater than the actual value [B1] (because the average magnitude of g is less than 9.81 m s^{-2}

Comments:
Stating that h is not small alone is not good enough

(ii) KE = ½ mv^2 where v = $2\pi r/T$ [M1]

v = 2 x π x (6800 + 6400) x 10^3/8.5 x 10^3 = 9.76 x 10^3 m s^{-1}

KE = ½ x 1500 x (9.76 x 10^3)^2 = 7.1 x 10^{10} J [A1]

(0 marks for method if use $\left(\frac{GMm}{2r}\right)$ as mass of Earth not given)

(iii) A geostationary satellite stays above the same point with respect to the earth and as such can be used for radio communications. [B1]
The satellite is not in geostationary orbit because its period is less than 1 day or 8.6×10^4 s. [B1]

Comments:
Many students merely stated what geostationary meant without discussing the advantage. Some students mentioned an advantage eg clearer images without explaining how it is linked to geostationary orbits.
Merely saying the period is different without evidence eg calculation or at least stating which value is larger is insufficient as it constitutes to guesswork.

6(a) (i) When the slide contact S is moved all the way up to the top of the 20 kΩ